

Problem Set 4

Problem 1: Do exercise 2.5.3 of the text.

Solution: The problem asks us to get the equations of motion using the Hamiltonian method for the system shown in Figure 1.5 (p. 46) of the text. Using x_1 and x_2 shown there as the generalized coordinates, the kinetic energy and potential energy are

$$T = \frac{m}{2}\dot{x}_1^2 + \frac{m}{2}\dot{x}_2^2, \quad V = \frac{k}{2}x_1^2 + \frac{k}{2}(x_1 - x_2)^2 + \frac{k}{2}x_2^2,$$

so the Lagrangian is $\mathcal{L} = T - V$ and thus the generalized momenta are

$$\begin{aligned} p_1 &= \frac{\partial \mathcal{L}}{\partial \dot{x}_1} = \frac{\partial T}{\partial \dot{x}_1} = m\dot{x}_1, & \Rightarrow & \dot{x}_1 = \frac{p_1}{m}, \\ p_2 &= \frac{\partial \mathcal{L}}{\partial \dot{x}_2} = \frac{\partial T}{\partial \dot{x}_2} = m\dot{x}_2, & \Rightarrow & \dot{x}_2 = \frac{p_2}{m}, \end{aligned}$$

so the Hamiltonian is

$$\mathcal{H} = T + V = \frac{p_1^2}{2m} + \frac{p_2^2}{2m} + \frac{k}{2}x_1^2 + \frac{k}{2}(x_1 - x_2)^2 + \frac{k}{2}x_2^2,$$

and Hamilton's equations are

$$\begin{aligned} \dot{x}_1 &= \frac{\partial \mathcal{H}}{\partial p_1} = \frac{p_1}{m}, & \dot{p}_1 &= -\frac{\partial \mathcal{H}}{\partial x_1} = -2kx_1 + kx_2, \\ \dot{x}_2 &= \frac{\partial \mathcal{H}}{\partial p_2} = \frac{p_2}{m}, & \dot{p}_2 &= -\frac{\partial \mathcal{H}}{\partial x_2} = -2kx_2 + kx_1. \end{aligned}$$

Problem 2: Do exercise 2.7.2 of the text.

Solution: (i):

$$\begin{aligned} \{q_i, q_j\} &:= \sum_k \left(\frac{\partial q_i}{\partial q_k} \cdot \frac{\partial q_j}{\partial p_k} - \frac{\partial q_i}{\partial p_k} \cdot \frac{\partial q_j}{\partial q_k} \right) = \sum_k \left(\frac{\partial q_i}{\partial q_k} \cdot 0 - 0 \cdot \frac{\partial q_j}{\partial q_k} \right) = 0 \\ \{p_i, p_j\} &:= \sum_k \left(\frac{\partial p_i}{\partial q_k} \cdot \frac{\partial p_j}{\partial p_k} - \frac{\partial p_i}{\partial p_k} \cdot \frac{\partial p_j}{\partial q_k} \right) = \sum_k \left(0 \cdot \frac{\partial p_j}{\partial p_k} - \frac{\partial p_i}{\partial p_k} \cdot 0 \right) = 0 \\ \{q_i, p_j\} &:= \sum_k \left(\frac{\partial q_i}{\partial q_k} \cdot \frac{\partial p_j}{\partial p_k} - \frac{\partial q_i}{\partial p_k} \cdot \frac{\partial p_j}{\partial q_k} \right) = \sum_k (\delta_{ik} \delta_{jk} - 0 \cdot 0) = \delta_{ij}, \end{aligned}$$

and

$$\begin{aligned} \{q_i, \mathcal{H}\} &:= \sum_k \left(\frac{\partial q_i}{\partial q_k} \cdot \frac{\partial \mathcal{H}}{\partial p_k} - \frac{\partial q_i}{\partial p_k} \cdot \frac{\partial \mathcal{H}}{\partial q_k} \right) = \sum_k \left(\delta_{ik} \cdot \frac{\partial \mathcal{H}}{\partial p_k} - 0 \cdot \frac{\partial \mathcal{H}}{\partial q_k} \right) = \frac{\partial \mathcal{H}}{\partial p_i} = \dot{q}_i, \\ \{p_i, \mathcal{H}\} &:= \sum_k \left(\frac{\partial p_i}{\partial q_k} \cdot \frac{\partial \mathcal{H}}{\partial p_k} - \frac{\partial p_i}{\partial p_k} \cdot \frac{\partial \mathcal{H}}{\partial q_k} \right) = \sum_k \left(0 \cdot \frac{\partial \mathcal{H}}{\partial p_k} - \delta_{ik} \cdot \frac{\partial \mathcal{H}}{\partial q_k} \right) = -\frac{\partial \mathcal{H}}{\partial q_i} = \dot{p}_i, \end{aligned}$$

where in the last steps I used Hamilton's equations.

(ii): The Hamiltonian given is $\mathcal{H} = p_x^2 + p_y^2 + ax^2 + by^2$. If $a = b$, $\mathcal{H}$ has a

symmetry under simultaneous rotations in the x - y and p_x - p_y planes, under which ℓ_z (the generator) is conserved. Therefore $\{\ell_z, \mathcal{H}\} = 0$.

We check this as follows:

$$\{\ell_z, \mathcal{H}\} = \sum_k \left(\frac{\partial \ell_z}{\partial q_k} \cdot \frac{\partial \mathcal{H}}{\partial p_k} - \frac{\partial \ell_z}{\partial p_k} \cdot \frac{\partial \mathcal{H}}{\partial q_k} \right) = \frac{\partial \ell_z}{\partial x} \cdot \frac{\partial \mathcal{H}}{\partial p_x} + \frac{\partial \ell_z}{\partial y} \cdot \frac{\partial \mathcal{H}}{\partial p_y} - \frac{\partial \ell_z}{\partial p_x} \cdot \frac{\partial \mathcal{H}}{\partial x} - \frac{\partial \ell_z}{\partial p_y} \cdot \frac{\partial \mathcal{H}}{\partial y}.$$

But

$$\begin{aligned} \frac{\partial \mathcal{H}}{\partial p_k} &= 2p_k, & \frac{\partial \mathcal{H}}{\partial x_k} &= \left(\frac{\partial \mathcal{H}}{\partial x}, \frac{\partial \mathcal{H}}{\partial y} \right) = (2ax, 2by), \\ \frac{\partial \ell_z}{\partial p_k} &= \frac{\partial(xp_y - yp_x)}{\partial p_k} = \left(\frac{\partial \ell_z}{\partial p_x}, \frac{\partial \ell_z}{\partial p_y} \right) = (-y, x), & \frac{\partial \ell_z}{\partial q_k} &= \left(\frac{\partial \ell_z}{\partial x}, \frac{\partial \ell_z}{\partial y} \right) = (p_y, -p_x), \end{aligned}$$

so

$$\{\ell_z, \mathcal{H}\} = p_y \cdot 2p_x + (-p_x) \cdot 2p_y - (-y) \cdot 2ax - x \cdot 2by = 2xy(a - b)$$

which vanishes if $a = b$.

Problem 3: Do exercise 2.8.1 of the text.

Solution: Since $g = p_1 + p_2$, it generates the infinitesimal transformations

$$\begin{aligned} \delta x_1 &= +\epsilon \frac{\partial g}{\partial p_1} = +\epsilon, & \delta p_1 &= -\epsilon \frac{\partial g}{\partial x_1} = 0, \\ \delta x_2 &= +\epsilon \frac{\partial g}{\partial p_2} = +\epsilon, & \delta p_2 &= -\epsilon \frac{\partial g}{\partial x_2} = 0. \end{aligned}$$

So, to order ϵ , these give the canonical transformations $x_i \rightarrow \bar{x}_i(x_j, p_j)$ and $p_i \rightarrow \bar{p}_i(x_j, p_j)$ with

$$\begin{aligned} \bar{x}_1 &= x_1 + \epsilon, & \bar{p}_1 &= p_1, \\ \bar{x}_2 &= x_2 + \epsilon, & \bar{p}_2 &= p_2, \end{aligned}$$

which is precisely a spatial translation of the whole system by an amount ϵ .